

PRODUCT DATA EXPLORATION USING EXCEL

Submitted By:

Usharani

Course:

Data Analytics

1. Introduction

This project is created using Microsoft Excel to analyse product data. Different Excel formulas and functions are used to perform calculations, logical operations, conditional analysis, and text formatting.

2. Dataset Description

The dataset contains the following columns:

Column Name	Description
Product ID	Unique ID of the product
Day	Day extracted from Product ID
Country Code	Country abbreviation
Month	Month extracted from Product ID
Product Name	Name of the product
Brand Name	Brand of the product
Price (\$)	Product price
Quantity	Number of products
Category	Product category
Price Range	Product classification

3. Functions Used in the Project

A. Basic Functions

1. SUM()

Purpose: Calculate the total price of all products.

Formula:

=SUM(G2:G35)

Result: 10,100

2. COUNTA()

Purpose: Count total number of products.

Formula:

=COUNTA(E2:E35)

Result: 34

3. AVERAGE()

Purpose: Find the average product price.

Formula:

=AVERAGE(G2:G35)

Result: 297

4. Minimum and Maximum Functions

1. MIN()

Purpose: Find the minimum product price.

Formula:

=MIN(G2:G35)

Result: 30

2. MAX()

Purpose: Find the maximum product price.

Formula:

=MAX(G2:G35)

Result: 1000

5. Logical Function

IF()

Purpose: Classify products as High Price or Standard Price.

Formula:

=IF(G2>=500,"High price", "Standard price")

Result: High price / Standard price

6. Conditional Functions

1. SUMIF()

Purpose: Calculate total price of Electronics products.

Formula:

=SUMIF(I2:I35,"Electronics",G2:G35)

Result: 8,050

2. COUNTIF()

Purpose: Count products with price less than \$100.

Formula:

=COUNTIF(G2:G35,"<100")

Result: 11

7. Text Functions

1. LEFT()

Purpose: Extract Day from Product ID.

Formula:

=LEFT(A2,2)

Result: 28

2. RIGHT()

Purpose: Extract Country Code from Product ID.

Formula:

=RIGHT(A2,2)

Result: US

3. MID()

Purpose: Extract Month from Product ID.

Formula:

=MID(A2,4,3)

Result: JAN

8. Conclusion

This project helped in understanding Excel formulas, logical functions, conditional functions, and text functions. It improved data analysis and formatting skills using Microsoft Excel.

Product ID	Day	Country Code	Month	Product Name	Brand Name	Price (\$)	Quantity	Category	Price Range
28-JAN-US	28	US	JAN	Laptop	Dell	1000	30	Electronics	High price
15-FEB-US	15	US	FEB	Sneakers	Nike	80	15	Fashion	Standard price
03-MAR-US	03	US	MAR	Coffee Maker	Keurig	130	40	Kitchen	Standard price
11-APR-US	11	US	APR	Smartphone	Samsung	800	25	Electronics	High price
22-MAY-US	22	US	MAY	Backpack	North Face	70	20	Outdoor	Standard price
07-JUN-UK	07	UK	JUN	Headphones	Sony	200	45	Electronics	Standard price
19-JUL-UK	19	UK	JUL	T-shirt	Adidas	30	5	Fashion	Standard price
23-AUG-UK	23	UK	AUG	Blender	Ninja	90	35	Kitchen	Standard price
05-SEP-UK	05	UK	SEP	Tablet	Apple	500	50	Electronics	High price
14-OCT-UK	14	UK	OCT	Hiking Boots	Timberland	130	10	Outdoor	Standard price
17-JUN-IN	17	IN	JUN	Laptop	HP	950	25	Electronics	High price
25-NOV-AU	25	AU	NOV	Sneakers	Adidas	90	40	Fashion	Standard price
08-DEC-DE	08	DE	DEC	Coffee Maker	Nespresso	120	35	Kitchen	Standard price
18-FEB-CA	18	CA	FEB	Smartwatch	Fitbit	150	15	Electronics	Standard price
16-APR-ES	16	ES	APR	Headphones	Bose	250	20	Electronics	Standard price
21-AUG-CA	21	CA	AUG	Laptop Bag	Samsonite	50	35	Accessories	Standard price
20-AUG-CN	20	CN	AUG	Smartwatch	Huawei	160	15	Electronics	Standard price
27-JAN-IT	27	IT	JAN	Laptop	Asus	980	10	Electronics	High price
04-MAR-UK	04	UK	MAR	Sunglasses	Oakley	150	15	Fashion	Standard price
14-AUG-US	14	US	AUG	Camping Tent	Coleman	200	10	Outdoor	Standard price
14-MAY-RU	14	RU	MAY	Camera	Nikon	700	50	Electronics	High price
09-JAN-CA	09	CA	JAN	Microwave	Panasonic	80	20	Kitchen	Standard price
19-JUL-BR	19	BR	JUL	Fitness Tracker	Xiaomi	150	30	Electronics	Standard price
21-AUG-CA	21	CA	AUG	Laptop Bag	Samsonite	50	35	Accessories	Standard price
28-SEP-CA	28	CA	SEP	Smartphone	Google	800	45	Electronics	High price
03-JUN-CA	03	CA	JUN	Sunglasses	Ray-Ban	130	25	Fashion	Standard price
11-JUL-CA	11	CA	JUL	Blender	Vitamix	400	40	Kitchen	Standard price
16-APR-ES	16	ES	APR	Headphones	Bose	230	20	Electronics	Standard price
07-MAR-CA	07	CA	MAR	Dress	Zara	60	30	Fashion	Standard price
13-APR-CA	13	CA	APR	Toaster	Hamilton	40	10	Kitchen	Standard price
24-MAY-CA	24	CA	MAY	Fitness Tracker	Garmin	130	5	Electronics	Standard price
02-DEC-CA	02	CA	DEC	Jeans	Levi's	50	50	Fashion	Standard price
17-JUN-RU	17	IN	JUN	Laptop	HP	950	25	Electronics	High price
09-JUL-FR	09	FR	JUL	Watch	Casio	100	20	Accessories	Standard price

QUESTIONS & ANSWERS

1. Sum, Count, and Average

Function	Purpose	Result
SUM()	Total price of all products	10,100
COUNTA()	Count total products	34
AVERAGE()	Average product price	297

2. Minimum and Maximum Values

Function	Purpose	Result
MIN()	Minimum price among all products	30
MAX()	Maximum price among all products	1,000

3. Logical Function – IF

Function	Purpose	Result
IF()	Classify products as High Price based on price value	High price
IF()	Classify products as Standard Price based on price value	Standard Price

4. Conditional Functions – SUMIF and COUNTIF

Function	Purpose	Result
SUMIF()	Total price of products in the Electronics category	8,050
COUNTIF()	Number of products with a price less than \$100	11

5. Text Formatting – LEFT, RIGHT, MID

Function	Purpose	Result
LEFT()	Create Day column using LEFT	28
RIGHT()	Create Country Code column using RIGHT	US
MID()	Create Month column using MID	JAN